

SUMMARY

Areas of expertise: Trustable and Explainable AI, Medical AI and Machine Learning applications.

Research:

- 60+ publications with a total of 2650 citations and an H-index of 25 (Google Scholar).
 - 9 high impact factor journals (including TPAMI, TMI, and TIP).
 - 15 CORE A/A* conferences (including CVPR, ICCV, and MICCAI).
- Supervising 8 PhD students, taking principal supervisor duty for 5 PhD students (since 2023), co-supervising 3 PhD students.
- Supervised 3 AI postdocs since 2021.
- Inventor on 2 US patents.

Teaching and supervision:

- Teaching Artificial Intelligence and Computer Vision courses at the School of Computer Science and Information Technologies, Adelaide University.
- Group project supervisor (UG/PG) for the School of CMS, supervised >40 Master's/Honours students since 2020.

Academic Services:

- Reviewing for conferences and journals: CVPR, AAAI, ICLR, ECCV, MICCAI, PR, TPAMI, TMI, PLOS.
- PC member for AusDM 2024. Local arrangement chair for DICTA 2025.

WORK EXPERIENCE/PROJECTS

Senior Lecturer

School of Computer Science and Information Technology, Adelaide University

Jan. 2026 — current

Adelaide, Australia

Senior Lecturer

School of Computer and Mathematical Sciences, University of Adelaide

Jan. 2025 — Dec. 2025

Adelaide, Australia

**Senior Research Fellow, Centre for Augmented Reasoning (CAR)
Postdoctoral Research Fellow**

Australian Institute for Machine Learning, University of Adelaide

Dec. 2021 — Dec. 2024

Jun. 2019 — Dec. 2021

Adelaide, Australia

- AI lead on the NHMRC partnership Cardiovascular Chest Pain analysis project (collaboration with Flinders Medical Centre) (official project page).
- Lead of the SA Health collaboration on digital pathology (Chronic Myelomonocytic Leukemia) and healthcare improvement. (news link)
- Lead of an international Orthopedics AI fracture recognition, detection and segmentation (collaboration with Flinders Medical Centre and the University of Groningen (UMCG), Netherlands).
- Medical domain visual question answering - Winner of the ImageCLEF VQA-Med and VQG-Med challenge (challenge leader board link).
- Synthetic data privacy and utility evaluation for Digital Health SA.

Postdoctoral Research Fellow

Dept. of Elec. and Comp. Eng., University of British Columbia

Nov. 2017 — May. 2019

Vancouver, Canada

- Machine learning based 2D echo cardiac image analysis including echo view classification, cardiac phase detection, left ventricular segmentation, ejection fraction estimation, echo image quality uncertainty analysis (Collaboration with Vancouver General Hospital).
- Aeroplane material crack imaging (Collaboration with Boeing).

FUNDING (>\$7M AUD AS A CHIEF INVESTIGATOR)

MRFF Early to Mid-Career Researchers, 2025, MRFF

\$4.4M

- (Chief investigator) Project Name: A novel AI-augmented digital system for medication deprescribing in residential aged care: the AiCT-Med national cluster randomised controlled trial. Led by Dr Nasir Wabe @ Macquarie University.

Round 5 National Industry PhD Award

1 PhD Scholarship

- (Chief investigator) Joint application with SA Pathology, \$40K industry contribution.

MRFF National Critical Research Infrastructure, 2023, MRFF

\$2.9M

- (Chief investigator) Project Name: AutoMedic: A scalable, smart solution to detect and resolve medicine harm.

Advancing Diagnosis or Treatment of Child and Adolescent Rheumatic Diseases, 2023, The Hospital Research Foundation.

\$125K

- (Chief investigator A) Project Name: Advancing the imaging diagnosis of juvenile idiopathic arthritis with deep learning.

Robinson Research Institute Innovation Seed Funding, 2023, University of Adelaide.	\$50K
• (Chief investigator) A predictive model for IVF cycle outcomes from personalised granulosa cell profile. Collaboration with	
Faculty of Health and Medical Sciences Research Infrastructure Funding Awards 2022, University of Adelaide	\$25K
• (Chief investigator) Project Name: A local and high-performance computational workstation for medical machine learning research.	
DIGI + FAME Strategy Internal Grant 2021, University of Adelaide	\$20K
• (Chief investigator) Project Name: Accelerating Drug Discovery with Machine Learning.	
Orthopedics AI, Collaborative Research Agreement, University of Groningen, The Netherlands	\$290K (2022-23) / \$150K (2024)
• (Chief investigator) Project Name: Orthopedics AI Research Collaboration.	
SA Pathology Collaboration Research	
• (Chief investigator) Project Name: Chronic myelomonocytic leukemia (CMML) morphological AI analysis.	

EDUCATION

PhD of Comp. Sci., Australian Centre for Robotic Vision, University of Adelaide	Aug. 2013 — Sept. 2017
Hons. of Comp. Sci. (1st Class), University of Adelaide	Aug. 2012 — Aug. 2013
B. of Comp. Sci., University of Adelaide, GPA: 6.375/7.0	Mar. 2009 — Aug. 2012

PATENTS

- Abolmaesumi, P., [Liao, Z.](#), Tsang, T. & Behnami, D. *Neural network image analysis* US Patent App. 17/393,955. Nov. 2021.
- Abolmaesumi, P., Rohling, R., Tsang, T., [Liao, Z.](#) & Abdi, A. *Ultrasonic image analysis* US Patent 10,751,029. Aug. 2020.

RELATED PUBLICATIONS

- Chowdhury, T. F., Huy, T. D., Pan, S., Stoddard, J. & [Liao, Z.](#) *AR&D: A Framework for Retrieving and Describing Concepts for Interpreting AudioLLMs in IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) (oral, to appear) (2026).*
- Chowdhury, T., Phan, V. M. H., Liao, K., Dong, N., To, M.-S., van den Hengel, A., Verjans, J. W. & [Liao, Z.](#) *Looking in the Mirror: A Faithful Counterfactual Explanation Method for Interpreting Deep Image Classification Models in Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) (CORE A*) (2025), 2239–2249.*
- Chowdhury, T. F., Liao, K., Phan, V. M. H., To, M.-S., Xie, Y., Hung, K., Ross, D., van den Hengel, A., Verjans, J. W. & [Liao, Z.](#) *CAPE: CAM as a Probabilistic Ensemble for Enhanced DNN Interpretation in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (CORE A*) (2024), 11072–11081.*
- Chowdhury, T. F., Phan, V. M. H., Liao, K., To, M.-S., Xie, Y., van den Hengel, A., Verjans, J. W. & [Liao, Z.](#) *AdaCBM: An Adaptive Concept Bottleneck Model for Explainable and Accurate Diagnosis in International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI) (CORE A) (2024), 35–45.*
- [Liao, Z.](#), Liao, K., Shen, H., Van Boxel, M. F., Prijs, J., Jaarsma, R. L., Doornberg, J. N., Van den Hengel, A. & Verjans, J. W. *CNN Attention Guidance for Improved Orthopedics Radiographic Fracture Classification. IEEE Journal of Biomedical and Health Informatics (JBHI) (IF: 7.7) (2022).*